VIA University College



Programming 3

Autumn 2026

Learning Objectives

- ✓ explain the basics of the **C4 Model,** for visualizing software architecture.
- ✓ give an overview of the diagrams that make up the C4 model

The C4 Model

Credits

Simon Brown (@simonbrown)
http://www.codingthearchitecture.com/

https://c4model.com/



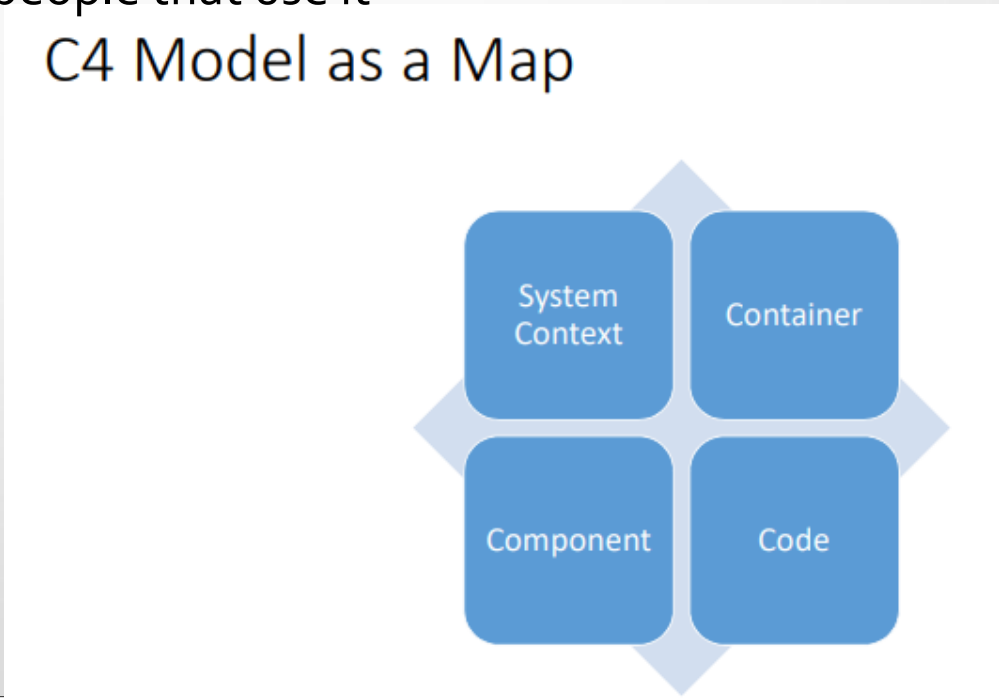
https://leanpub.com
/visualising-software-architecture
/c/HaarlemSoftwareDevelopmentMeetup

Architectural Styles - Examples

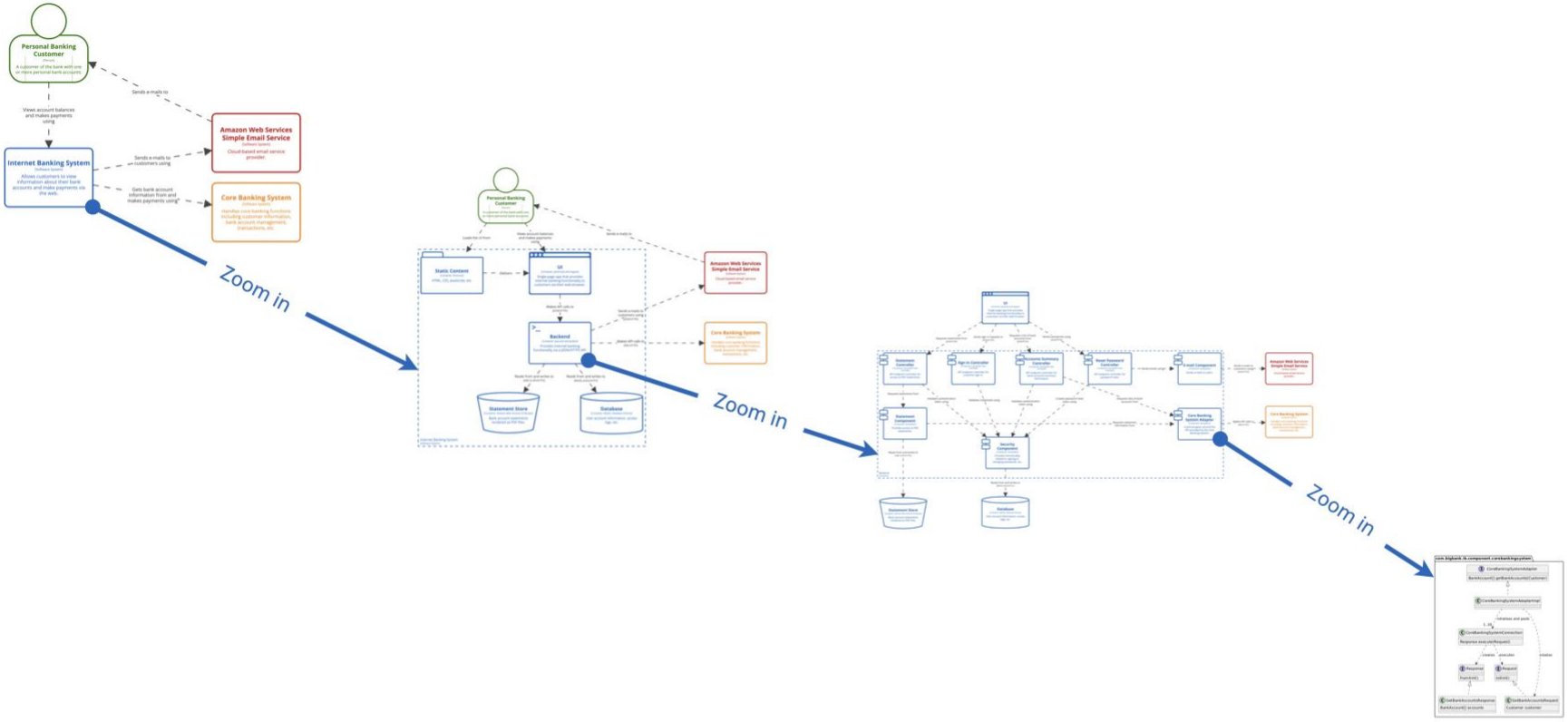
- Peer-To-Peer
- Three-tier, N-Tier Architecture
- Microservices

Diagramming and Documenting Software Architecture

- The main goal of the C4 model is to simplify architecture communication.
- Static modeling of:
  - Software system for the people that use it
  - Containers
  - Components
  - Code



Static Structure Diagrams



System Context | Containers | Components | Code

Some Definitions



- **Person** – any user of the system
- **Software system** - Something that delivers value to its users
  - System you are modeling AND systems on which your software depends
- **Container** – something that represents an application or a data store
  - Something that needs to be running for the overall software system to work
  - Communication between containers typically takes the form of an inter-process communication.
- **Component** – describe internal structure of app containers, represents a collection of code; grouping of related functionality
- **Code** – represents internals of components based on the underlying programming language (eg. Clasess, interfaces, etc). Structure of code in a a single component

Example Containers

- App Container
  - Server-side web application, Client-side web application, Client-side desktop application, Mobile app, Server-side console application
  - Microservice: A single microservice, hosted in anything from a traditional web server to something like Spring Boot, Dropwizard, etc.
  - Serverless function: A single serverless function (e.g. Amazon Lambda, Azure Function, etc).
  - Shell script
- Data store container
  - Database (eg. PostgreSQL)
  - File system: A full local file system or a portion of a larger networked file system
  - Blob or content store: A blob store (e.g. Amazon S3, Microsoft Azure Blob Storage, etc) or content delivery network (e.g. Akamai, Amazon CloudFront, etc).

Core Diagrams

System Context Diagram

- Context diagram
  - Highlights the **software system** that we are describing
  - Focus on **people** (actors, roles, personas, Bob, Wendy) who use it and
  - the other software systems that interact with it
- No mention of technologies, protocols and other low level details.
- Zoomed out view – showing the big picture

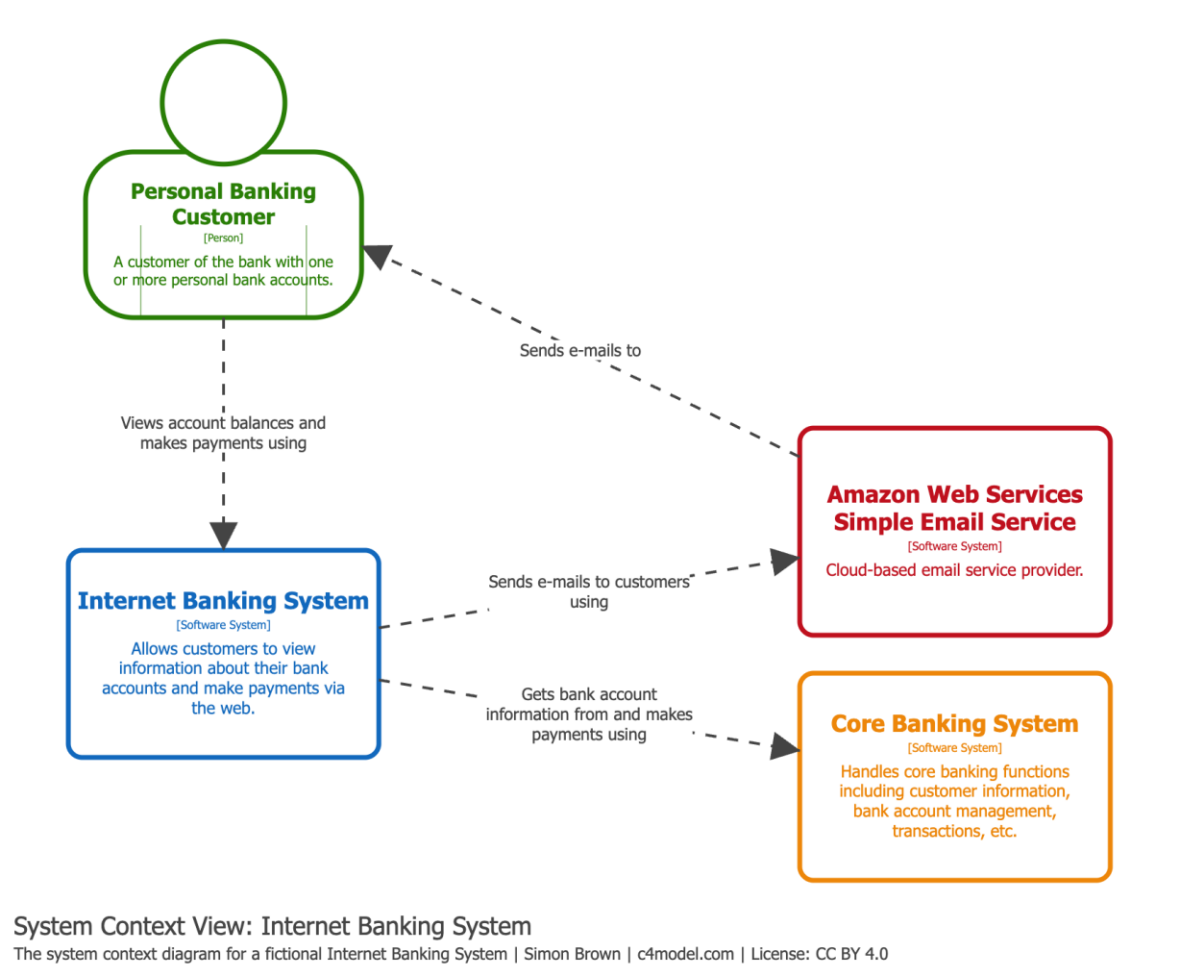
System Context Diagram

- Zoomed out view – showing the big picture
- Context diagram (multiple versions – for admin, user,…)
  - Highlights the **software system** that we are describing
  - Focus on **people** (actors, roles, personas, Bob, Wendy) who use it and
  - the other software systems that interact with it



- Intent:
  - What is the software system that we are building?
  - Who is is using the software system?
  - What are the various types of users doing with the software system?
  - How does the software system fit into the existing system landscape?

System Context Diagram - Example

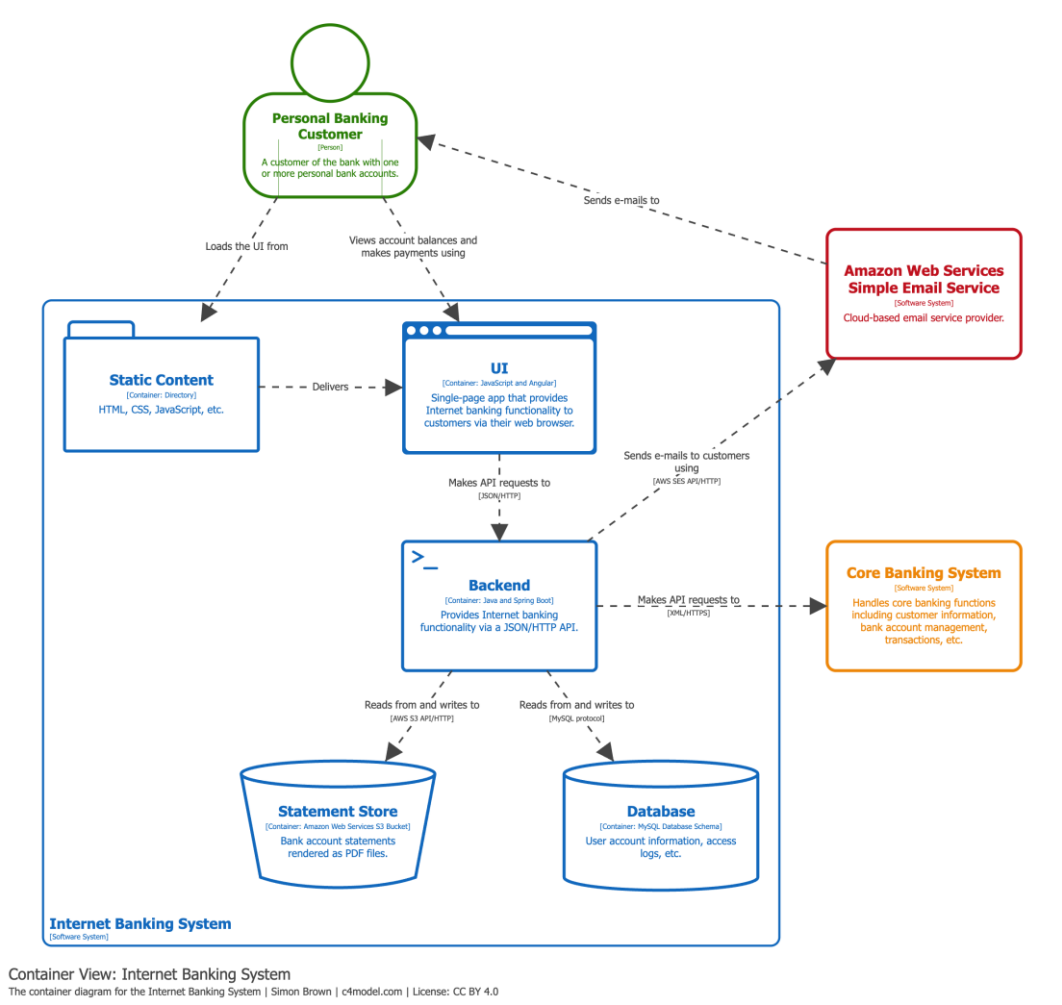


https://www.archi-lab.io/infopages/architecture/c4-model.html

Container Diagram

- Describe application and data stores that exist within the software system boundary (e.g. databases, APIs, microservices, etc).
- Intent:
  - How has the software system been decomposed into applications and data stores?
  - What are the responsibilities of those applications and data stores?
  - What are the primary technology choices (eg. Programming languages and products) used to implement those applications and data stores?
  - How do the applications and data stores communicate with one another?
  - As a software engineer, where do i need to write code to add a feature?

Container Diagram

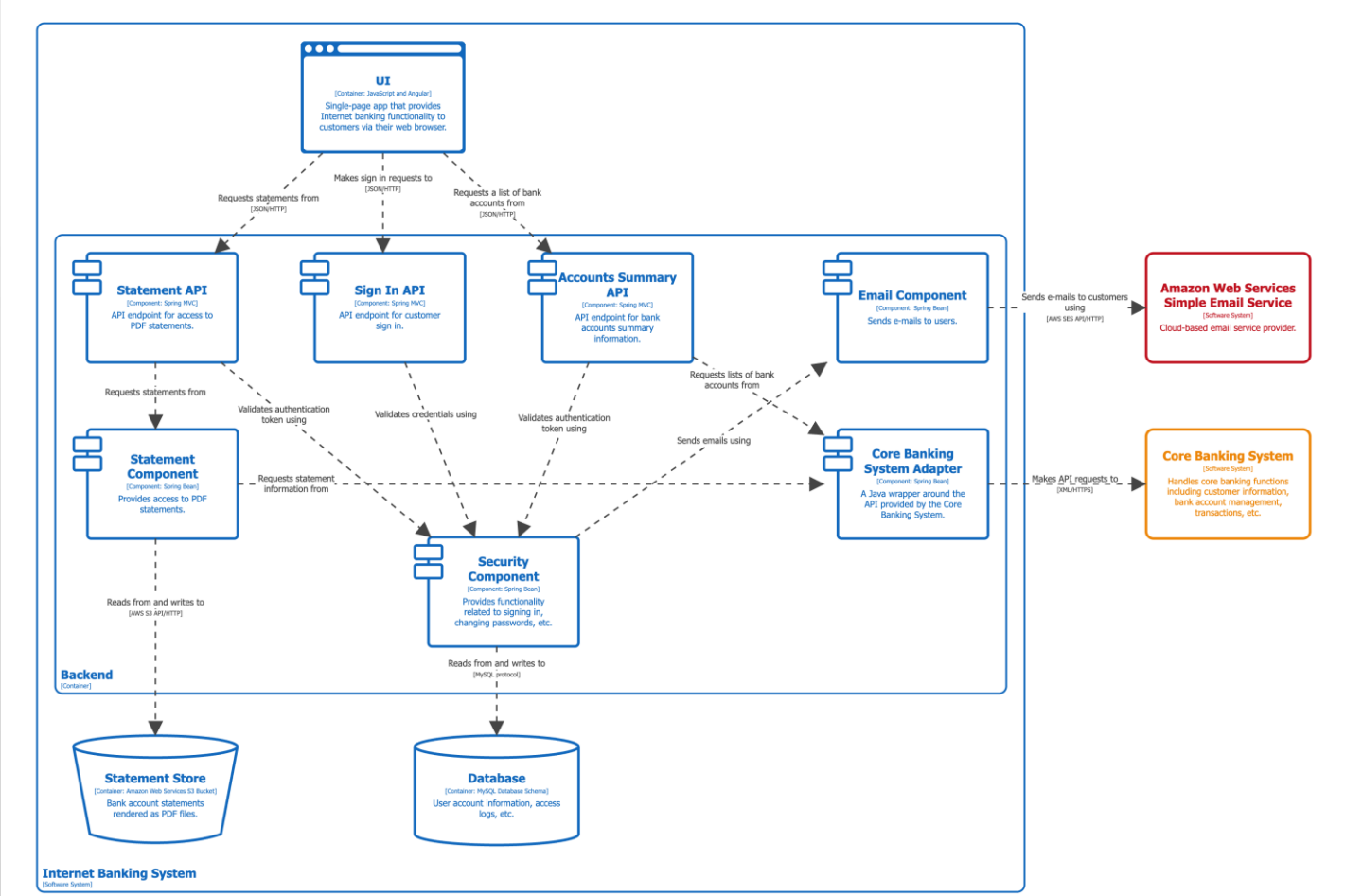


https://www.archi-lab.io/infopages/architecture/c4-model.html

Component Diagram

- Provides a way to zoom in to show the components that reside inside a single container. (e.g. controllers, repositories, queues, etc.)
- Intent:
  - How has the container been decomposed into a collection of componenets?
  - What are the responsibilities of those components?
  - How do the components collaborate and communicate with one another?
  - What are the primary technology choices (e.g. frameworks and libraries, or whatever you know) that will be used to implement the components?
  - As a softrware engineer, where do i need to write code to add a feature?

Component Diagram



Component View: Internet Banking System - Backend

The component diagram for the Internet Banking System Backend | Simon Brown | c4model.com | License: CC BY 4.0

Code Diagram

- Provides a way to show the internal implementation details of a component. Illustrates the structure of the code within a single component
- Intent:
  - How is a component decomposed into code-level building blocks?
  - Which of these code level building blocks are publicly exported/exposed by the component and which are internal implementation details?
  - How large and complicated is the component?

Focus on Levels 1 - 3

- Most architecture communication happens at:
  - Context
  - Container
  - Component



Group Discussion



- Discuss two key points from the lesson so far, that could be useful in SEP3
- Discuss ideas for your architecture design